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SCIENCE
PASSION
TECHNOLOGY

Zero-Shot Formation Classification in Amateur American Football

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Outline

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Why automate formation tagging?

- American football is “chess on the field”: pre-snap formations carry tactical information coaches rely on
- At the amateur level (e.g. Austrian Football League), formation tagging on platforms like Hudl is done **entirely by hand**
- Small programs have no staff to dedicate to film review
- Open-weight Vision-Language Models (VLMs) can be run locally, could they help?

Research Questions

RQ1 To what extent can user-grade open-source VLMs identify offensive and defensive formations from pre-snap images in a zero-shot setting?

RQ2 How do Qwen-VL, Gemma, and LLaMA compare in their ability to detect formations and player alignments?

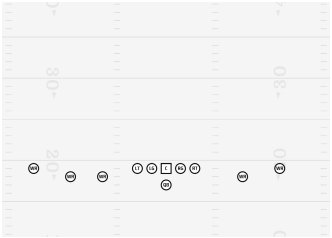
RQ3 What are the main failure modes and limitations when applying VLMs to amateur-quality football broadcast footage?

What a pre-snap formation shows

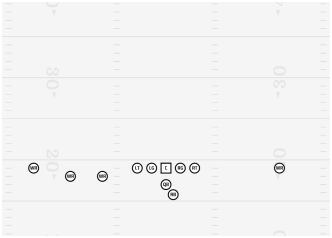
- Backfield occupied (RB/FB), or empty?
- How many wide receivers per side?
- Tight end/fullback on the line?
- Which side has the extra receiver (“strong side”)?

Why this is hard: precise counting and spatial reasoning

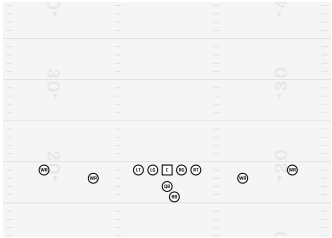
The Formation Classes (1/2)



EMPTY

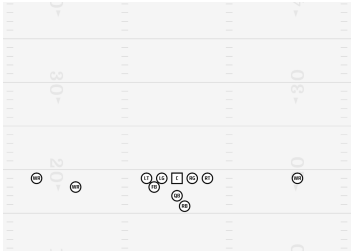


TRIO

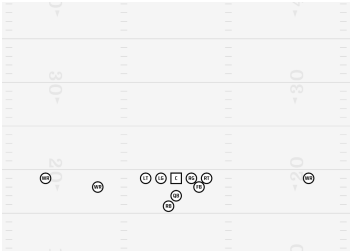


DOUBLES

The Formation Classes (2/2)



TREY



TANK

Dataset

- 40 offensive plays, 4 AFL games
- One pre-snap frame per play
- Manual annotations: formation, play type, jersey colors, offense side
- Majority class: **TRIO** (35%)

Formation	Count	%
TRIO	14	35.0
DOUBLES	11	27.5
EMPTY	5	12.5
TREY	5	12.5
TANK	5	12.5
Total	40	100.0

Models Under Test

Model	Params	Quantization	Disk size
Qwen3-VL:30b	30B	4-bit	19 GB
Gemma3:27b	27B	4-bit	17 GB
LLaMA3.2-Vision:11b	11B	4-bit	8 GB

- All served locally via **Ollama**, single 24 GB GPU
- Deterministic decoding (`temperature=0`), fully zero-shot: no fine-tuning, no few-shot examples

Evaluation Protocol

Phase A Text-only: do models know the formation *names* at all, with no image?

Phase B Basic image understanding: jersey colors, offense side, player counts, QB alignment

Phase C Formation classification: the core task, in four prompting strategies

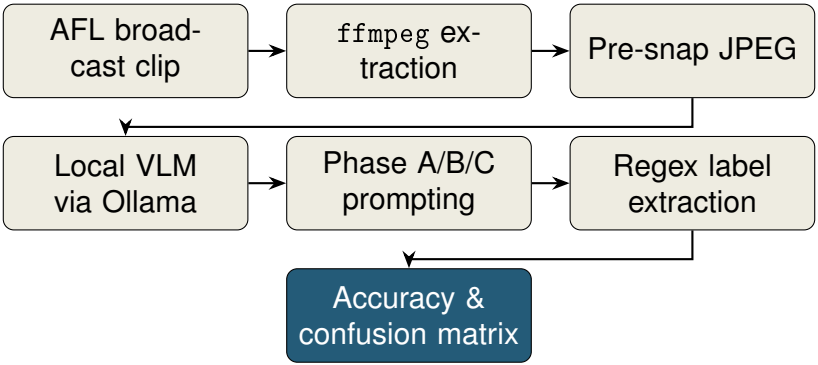
Majority-class baseline: always predict TRIO $\Rightarrow$ **35.0% accuracy**

Phase C: Four Prompting Strategies

C1	C2	C3	C4
Baseline: image + class names only	+ natural-language description per class	+ step-by-step decision tree	+ explicit vocabulary correction

C4 isolates one variable: does correcting the Phase A naming prior improve accuracy?

Pipeline Overview



Code Walkthrough

- Live in `evaluation.ipynb`: Ollama API call and prompts C1 $\rightarrow$ C4

Phase A: A Naming Bias, Before Any Image (1/2)

Q	Asked for	Qwen3-VL	Gemma3	LLaMA3.2
A2	EMPTY (no backs)	✓	✓	✓
A3	TANK, TREY (run game)	×	✓	×
A4	**TRIO** (3 WR one side)	×	×	×
A5	DOUBLES (2+2 WR)	✓	✓	✓

Phase A: A Naming Bias, Before Any Image (1/2)

Finding

All three models say “TREY” for three-receivers-one-side (correct: TRIO)

Phase B: Can Models Tell Offense from Defense?

- Sanity check: jersey colors, offense side, player counts, QB stance
- **Swap analysis**: of all offensive-jersey-color errors, how many are the model naming the *defense's* true color instead?
- Even with the offense side given explicitly (Phase B2), does the swap rate change?

Phase B: Swap Analysis Results

Model	Correct	Swapped	Other wrong
Qwen3-VL:30b	15	11	14
Gemma3:27b	11	19	10
LLaMA3.2:11b	15	21	4

"Swapped" = model named the defense's true color when asked for the offense's.

Swap rate among all errors: **65%** (side inferred) vs. **54%** (side given, B2): still the majority of errors even when told which side is offense.

Phase C: Formation Classification Accuracy

Model	C1	C2	C3	C4
Qwen3-VL:30b	10.0%	30.0%	27.5%	**37.5%**
Gemma3:27b	15.0%	**37.5%**	12.5%	32.5%
LLaMA3.2-Vision:11b	12.5%	12.5%	20.0%	**35.0%**
Majority baseline		**35.0%**		

C4: Testing the Naming-Prior Hypothesis

- same bare framing as C1, plus *only* an explicit correction of the TRIO/TREY naming confusion found in Phase A
- Isolates one variable: does correcting the prior improve accuracy, independent of extra descriptive content (C2) or reasoning structure (C3)?
- Qwen jumps 10.0% $\rightarrow$ 37.5%; LLaMA jumps to its best result (35.0%)

C4: What Actually Changed

Model	C1 top prediction	C4 top prediction
Qwen3-VL:30b	TREY (92.5% of images)	TRIO (95.0% of images)
Gemma3:27b	TREY (97.5% of images)	TRIO (95.0% of images)
LLaMA3.2-Vision:11b	TREY (100% of images)	TRIO (100% of images)

- C4 retargets the same single-label collapse from TREY (12.5%) onto TRIO (35%) $\rightarrow$ accuracy rises only because the new default guess matches the majority class.

Discussion (1/2)

- **A language prior can beat visual evidence.** A naming confusion learned during pretraining may persist even with the image in front of the model.
- **Team-role assignment may fail before semantics even start.** If models can't reliably tell offense from defense, formation logic is built on a shaky foundation regardless of vocabulary.

Discussion (2/2)

- **More structured prompting isn't guaranteed to help.** Descriptions and decision trees add information, but may not overwrite an internalized label mapping.
- **Scale doesn't clearly help.** Qwen (30B) and Gemma (27B) both peak at 37.5%; LLaMA (11B) is close behind at 35.0%.

Limitations

- Small test set: 40 images, minority classes have only 5 samples each
- Single competition (AFL): may not generalize to other leagues/cameras
- All models run at 4-bit quantization, constrained by 24 GB VRAM
- Purely zero-shot: no fine-tuning, no in-context examples

Conclusion RQ1 & RQ2

RQ1 Two conditions exceed the majority baseline (37.5% vs. 35.0%), but every condition tested collapses onto one dominant label: not evidence of genuine per-image classification.

RQ2 No model shows real multi-class discrimination. Peak accuracy doesn't track parameter count (30B and 27B tie at 37.5%, 11B close behind).

Conclusion RQ3

RQ3 Two compounding failure modes:

- A pretrained naming prior conflates this dataset's TRIO/TREY convention: confirmed causally by C4 re-collapsing onto TRIO instead of TREY
- Role-assignment fragility: most jersey-color errors (65%, still 54% with side given) are literal offense/defense swaps

Future Work

- Few-shot prompting with corrected examples
- Fine-tuning on a larger, multi-league dataset

Master's thesis

Computer-vision pipeline: player detection and bird's-eye-view transform of the field, then a **rule-based** classifier assigns the formation.

Thank you

Questions?